

Phuoc, Vo Duc



SUMMARY

- Has 1+ year of experience in flutter development (convert native project to flutter project).
- Has 2+ year of experience in MAUI development.
- Has 3+ years of experience in Android development.
- Has 1+ years of experience in .Net development.
- Has 6 months of experience in iOS development. (can do)
- Has 1+ year of experience in Wpf development.
- Has 7+ years of experience in Xamarin .Forms development.
- Interested in Android development, iOS development, Object Oriented Programming, data structures and algorithms, Computer graphics.
- Quick learner

COMPUTER SKILLS

- **Languages:**
 - C#, Python, C++ Java, Objective-C, Javascript.
 - Verilog, Assembly, C.
- **Operating Systems:**
 - Windows XP, 7, 8, 10.
 - Linux.
 - Mac OS X.
- **Development, Design Tools:**
 - Visual Studio, Android Studio, XCode, Notepad.
 - GIT, SVN, JIRA.
 - Photoshop, Paint.
- **Databases:**
 - RDBMS: SQL Server, SQLite.
 - No SQL: MongoDB.
 - Graph DB: Neo4j.
- **Other:**
 - OOP, Design Patterns, Design Database, Excel skill.

EDUCATIONAL BACKGROUND

- Da Nang University of Science and Technology - BS Major in Software engineering.
- Aptech Program ACCPI-10.
- Internship at FPT .Net position.
- Internship at AsianTech iOS position.

CONTACT



PROFESSIONAL EXPERIENCE

Advanced Technical Projects

Mini CPU

Project Description:

Designed a 32-bit Verilog CPU with ALU, registers, and RAM; implemented testbench verifying arithmetic, logic, and memory instructions.

AI Voice Assistant Project

- Developed an AI voice assistant using **ESP32** with microphone (INMP441) and speaker (MAX98357A).
- Built a **Python proxy server** to handle audio streaming, speech recognition, AI response processing, and text-to-speech.
- Integrated **Google Speech-to-Text** for voice recognition and **Google Text-to-Speech / Murf AI** for natural voice output.
- Implemented **WebSocket-based real-time audio communication** between ESP32 and server.
- Designed the system to offload heavy processing to the backend, enabling smooth performance on low-resource hardware.

Drone Control System

Role: Embedded Developer | IoT Engineer

Technologies: ESP32, Arduino (C++), PWM (12-bit), PID, BNO055, SIM A7680C (MQTT over Cellular)

Project Description:

Designed and implemented a quadcopter prototype featuring real-time flight stabilization and remote control via MQTT over a cellular network. The system focuses on PID-based motion control, motor calibration, and reliable IoT communication using a SIM-based module.

Video Demo:

<https://www.youtube.com/watch?v=dA98d5y-v7Y>

Key Contributions:

- Developed **PID control algorithms** to stabilize roll, pitch, and yaw using real-time data from the BNO055 sensor.
- Controlled **four brushless motors** via 12-bit, 100Hz PWM signals through ESCs for stable and responsive flight.
- Implemented a **motor calibration routine** to ensure balanced thrust and consistent takeoff behavior.
- Integrated **SIM A7680C module** to communicate with an MQTT broker over the cellular network for remote commands and telemetry.

Smart Wi-Fi Controlled Power Socket

- Designed and built a smart power socket using an ESP32-C3 microcontroller and a solid-state relay (SSR) to control electrical appliances remotely.
- Implemented a real-time WebSocket-based interface to control and monitor the socket from a custom-built web UI.
- Programmed firmware in C++ (Arduino framework), structured communication using JSON, and handled Wi-Fi connectivity with auto-reconnect.
- Focused on stability and expandability, enabling easy addition of scheduling, energy monitoring, or mobile app integration.
- Technologies used: ESP32-C3, WebSocket, SSR Relay, Arduino (C++), HTML/CSS/JS.

AI/ML

Natural Language Processing (NLP) Skills:

- Developed and fine-tuned sentiment analysis models for Vietnamese text using deep learning and transformer-based architectures (e.g., PhoBERT).
- Performed text preprocessing, data splitting (train/test), and customized training on self-collected datasets to improve model accuracy.
- Built inference pipelines outputting clear sentiment labels (negative, neutral, positive) with associated confidence scores.
- Continuously expanded training data to enhance model understanding of contextual nuances.
- Proficient with Hugging Face Transformers library and TensorFlow/PyTorch frameworks for NLP tasks.

Computer Vision (CV) Skills:

- Fine-tuned a pre-trained MobileNetV2(TensorFlow) model by adding a custom label (1001 classes total)
- Converted the trained model to TensorFlow Lite (.tflite) for mobile deployment
- Integrated the TFLite model into a Flutter app for real-time on-device image classification

2024/06-Now

MAUI Developer

Project: Fnet IOT (Eastgate software) Taxi app (Convert android native to MAUI).

Team Size: 5

Responsibilities:

- Init Firebase (real time database, push notification...)
- Link app iOS: <https://apps.apple.com/vn/app/hilfe-iot/id6504023639>

2024/03-2024/05

Wpf Developer

Project: Westfalia (Eastgate software) Desktop app (Draw diagram).

Team Size: 8

Responsibilities:

- Apply Dijkstra algorithm to find shortest router.

2023/12-2024/03

Flutter Developer

Project: Idriver (IconTech) Taxi app (Convert android native to Flutter).

Team Size: 1

Technologies Used: gRPC, mqtt, firebase.

Responsibilities:

- Generate .proto file to model .dart file. Include to Flutter project and use gRPC.
- Init Firebase (auth, chat, google map...)
- Link app android: <https://play.google.com/store/apps/details?id=com.iconsys.idriver>
- Link app iOS: <https://apps.apple.com/vn/app/idriver/id1642272913>

2022/03 – 2023/12

Android Developer

Project: Icaller, Idriver (IconTech) Taxi app(multi company).

Team Size: 15

Technologies Used: gRPC, mqtt, firebase.

Responsibilities:

- <https://play.google.com/store/apps/details?id=com.iconsys.icaller>
- <https://play.google.com/store/apps/details?id=com.iconsys.idriver>

2022/01 – 2022/03

Xamarin .Forms Developer

Project: SAS-Family, (RikkeiSoft), NEC VietNam

Team Size: 06

Responsibilities:

- Used BLE, read battery level of other dive on background, foreground mode
- Read Rssi to warning.

2021/07 – 2021/12

Xamarin .Forms Developer

Project: IcoCa, (FOIS), NEC VietNam

Team Size: 10

Responsibilities:

- Build core project (high quality), multi level code review.
- Create base page, base control.

2019/11 – 2021/06

Full stack Developer

Project: LaratioID

Link app:

Apple: <https://apps.apple.com/app/id1531670198>

Android: <https://play.google.com/store/apps/details?id=com.xamarin.laratioid>

Description: Build back end api, mobile app and admin tool.

Team Size: 1

2019/09 – 2019/11

Xamarin .Forms Developer

Project: Yamaha Passport, (VietIS), NRI

Project Description:

This application help user of Yamaha, register motorcycle, view service history, find dealer, message.

Domain: Business

Team Size: 12

Technologies Used: Android SDK, Xcode, web socket, ngrok, .Net core.

Responsibilities:

- Fix a lot of difficult bug.
- Used GPS, calculate distance.
- Used dependence service, messaging center, app life cycle to create common class help ask again all permission.
- Handle scrollview to make parallax scroll view.
- Used excel, import data from <https://countrycode.org/>, read SIM info to detect dialing.
- Engineer and related jobs (team management, proposal, design, coding, testing, etc.).
- Other jobs of training new members and developing business for VietIS.

2016/11 – 2017/01

Android Developer (DGSoft)

Project: Emiclib Phone

Project Description:

This is mobile application, it would allow end users to read books, watch videos, listen to audio, users can download ebook and store in their mobile. From application, users also can search book, choice and view on mobile screen. For social network, users can do rating, and sharing a book with their friends. Users also view rating report by graph view for more comfortable to make decision.

Domain: Business

Team Size: 1

Technologies Used: Android SDK, Eclipse, Social Framework (Facebook), RESTful Service, HTTPClient, DDMS tool, Fragment, WebView, Photoshop, other library for android.

Responsibilities:

- Worked with Backend team to analyze requirement about UI/UX, new features, interface contract between service and application.
- Investigated and resolved issues and defects from customer.
- Developed and debugged application on simulator (AVD Manager - Android Virtual Device Manager) and real device.
- Used some controls: Gridview, Listview, Scrollview, Fragment to display the layout view and data based on UI/UX design.
- Worked with jQuery to graph reports on Android Canvas view.
- Worked with Javascript, HTML, CSS and API JavascriptInterface to resolved show menu 3 layer on WebView.
- Implemented login, rating and sharing by using Social Network (Facebook and Twitter).
- Used HTTPClient to call RESTful service to create, read, update and delete data and handling JSON data response.
- Used FragmentManager for managing, adding, replacing, removing fragments dynamically.

2015/04 – 2016/11**.Net Developer****Project: EmicLib Web (DGSoft)****Project Description:**

This is web product application, it would allow end users to find book in an internal library or public library, view book, request borrow, rating and sharing a book quality on their Social Network. EmicLib web allow admin manage book in their library, Include: Import book, pricing book, approve request borrow book.

Domain: Business

Team Size: 3

Technologies Used: Visual Studio, Telerik, SQL Server, SVN, Excel, Word.

Responsibilities:

- Worked with team leader to receive requirement and fix bug.
- Used .Net API to import/export Excel/Word file.
- Used Telerik Control to coding.
- Research and used algorithms to coding.
- Installed product for customer.
- Integrated C# DLL in to VB project.
- Rewrite SQL command in SQL server stored.
- Debug SQL, debug code, debug javascript and fix bug.
- Fix bug UI (HTML, Javascript, jQuery, CSS).
- Integrated Z39.50.DLL to connect public Library Database (Library of Congress, National Library of Vietnam).

